

Bernoulli Random Variables



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Abstract

A procedure for generating Bernoulli random using a computer program is discussed. The dependence of the sample mean on the number of Bernoulli random variables it is computed from is then investigated. We show that the sample mean gets progressively closer and closer to the expectation for the random variable as the number of random variables it is computed from increases.

1 Introduction

The Bernoulli random variable is used to describe the results from experiments that, like the flipping of a coin, have two possible outcomes. We use a Bernoulli random variable to specify whether a particular thing happened when we did our experiment. For example, we can describe the outcome of a test to test for a disease using the Bernoulli random variable, X . This random variable will have $X = 1$ if the person has the disease or $X = 0$ if the person is healthy. We often call these sorts of true/false experiments trials and thus say that the parameter, p , for the Bernoulli random variable gives us the probability that a particular trial will be successful.

The probability mass function for the Bernoulli random variable is:

$$P(X = x) = \begin{cases} (1 - p) & \text{if } x = 0 \\ p & \text{if } x = 1 \\ 0 & \text{otherwise} \end{cases}$$

The function below generates Bernoulli random variables.

```
1 import numpy as np
2
3 # The argument to the function is the parameter for the random variable p.
4 def bernoulli(p) :
5     # A uniform random variable is used to write a proposition that has a
6     # probability p of being true
7     if np.random.uniform(0,1)<p : return 1
8     return 0
```

Listing 1: Generating Bernoulli random variables

In the following report, I will first show, in section 2 how one can use this function to generate and plot a sample of Bernoulli random variables. In section 3, I will then discuss the expectation of the Bernoulli random variable and will show that the sample mean converges to the expectation.

2 Generating a sample

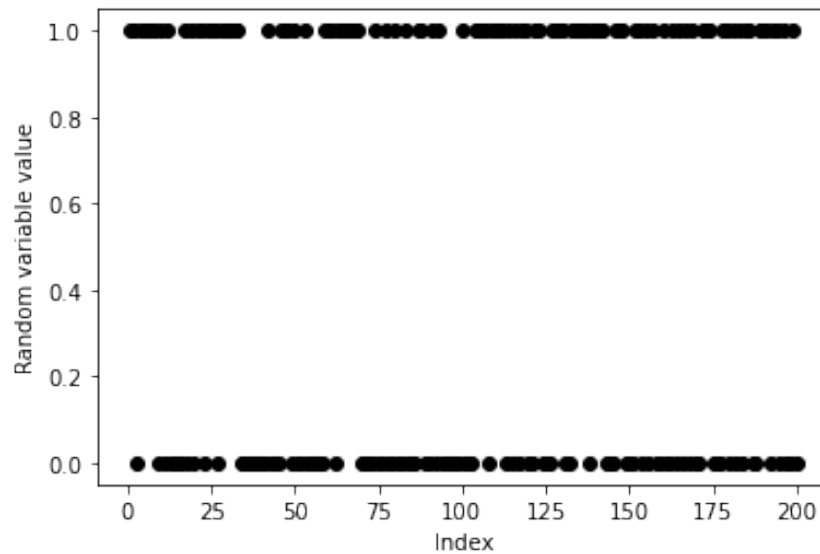


Fig. 1. The values of 200 Bernoulli random variables that were generated by repeatedly calling the function in the text.

I used the function for generating a Bernoulli random variable within a loop that generated the 200 Bernoulli random variables whose values are shown on the y-axis of the graph in figure 1. You can see from this graph that all the random variables are either zero or one, which is what would be expected given the probability mass function above.

3 Converging the average

The sample mean for a set of n random variables is calculated as:

$$\mu = \frac{1}{n} \sum_{i=1}^n X_i$$

Each of the X_i in this sum is an identically distributed random variable. In what follows, all the X_i values are independent Bernoulli random variables. I wrote some code to investigate how the value of μ depends on n . I know that for random variables of most types, the sample mean should converge on a number called the expectation, $\mathbb{E}(X)$, of the random variable. For a discrete random variable, the expectation is computed as:

$$\mathbb{E}(X) = \sum_{x=0}^{\infty} xP(X = x)$$

so for our Bernoulli random variable, we thus have:

$$\mathbb{E}(X) = 0.(1 - p) + 1.p = p$$

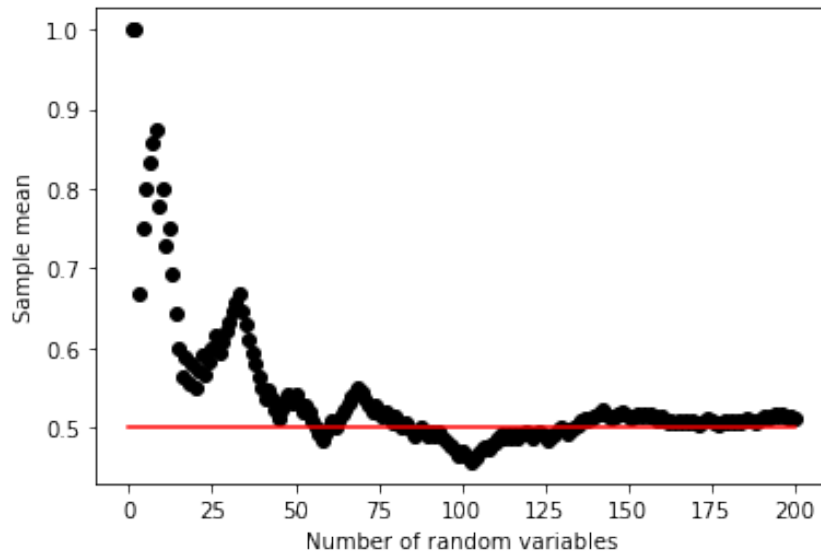


Fig. 2. The black dots in the figure above show the sample mean as a function of the number of Bernoulli random variables that it is computed from. The red line is the expectation for the random variable.

Figure 2 shows that sample mean does indeed converge onto the expectation of the value of this quantity. The black dots show the value of the sample mean calculated with different values of n . The true expectation for the random variable is then indicated using a red horizontal line. You can see in figure 2 that as n gets larger, the black dots get progressively closer and closer to this red line.